[redacted]

Dataset Details:

- 120 Images, 12 images for each digit 0-9
    - Training Set: 80 images
    - Validation Set: 20 images
    - Testing Set: 20 images
    - Resized to 28x28 pixels
    - Converted to grayscale
    - Normalized using MNIST statistics (mean=0.1307, std=0.3081)

Pre-trained model selection:

We selected ResNet18 as our pre-trained model for several key reasons:

- Efficient architecture with good performance-to-size ratio (60k images)
- Easily modifiable first layer for grayscale input
- Well-suited for transfer learning tasks

Hyperparameter Tuning Process:

Model 1:

- Froze all ResNet18 layers except final fully connected layer
- Training Parameters:
    - Batch size: 128
    - Learning rate: 0.01
    - Scheduler: StepLR
    - Epochs: 35

Model 2:

- Same as Model 1
- Training Parameters:
    - Batch size: 16
    - Learning rate: 0.001
    - Epochs: 40
    - No scheduler

Model 3:

- Unfroze layer4 and final layer

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- Training Parameters:
  - Batch size: 16
  - Learning rate: 0.001
  - Epochs: 40
  - No scheduler